

KRISHNA SHARMA

AI ML Engineer | Software Engineer

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PROFESSIONAL SUMMARY

AI/ML and software engineer experienced in building PyTorch computer-vision pipelines, RAG search, FastAPI services, and full-stack mobile workflows. Strong in model evaluation, API design, SQL, and turning technical prototypes into documented, testable applications.

CORE SKILLS

Languages: Python, SQL, JavaScript, TypeScript, Dart, C++, Java

AI and ML: PyTorch, scikit-learn, NumPy, Pandas, CNNs, Transformers, Computer Vision, Transfer Learning, Grad-CAM, NLP

Retrieval and APIs: RAG, Embeddings, BM25, Hybrid Retrieval, FastAPI, REST APIs, PostgreSQL, MySQL, SQLite

Frameworks and Tools: Git, Docker, Prometheus, React, Next.js, Flutter, Supabase, PostgreSQL

PROFESSIONAL EXPERIENCE

RentMyStay - BrightPath Technology & Services Pvt. Ltd.

Nov 2025 - Present

Full Stack AI Developer | Bengaluru, India

- Built FAQ-focused RAG search using hybrid retrieval that combines BM25 keyword matching with embedding similarity, exposed through FastAPI and SQL-backed workflows.
- Developed production API flows for AI search and structured data operations; configured Prometheus monitoring for 1,000+ daily requests, including latency, volume, and failure behavior.
- Maintained and debugged the Flutter app, integrated chat and notification APIs, and released a new iOS admin app.

Saral Startup Schools

Jan 2025 - Jul 2025

Software Engineering Intern | Bengaluru, India

- Built a responsive wedding-service booking platform with Next.js, authentication, dashboards, Drizzle ORM, and Cloudflare SQLite/D1 data flows.
- Designed the end-to-end browsing and booking journey across customer-facing pages and operational dashboard views.

PROJECTS

Coswara Respiratory Symptom Classifier | Python | PyTorch | Audio ML | [GitHub](#)

- Built a four-label CNN pipeline using mel spectrograms from 5,067 Coswara audio examples to predict fever, cold, fatigue, and cough.
- Addressed class imbalance with weighted focal loss and used mixup augmentation; trained and evaluated with five-fold cross-validation.
- Achieved **0.3900 ± 0.0135 mean best-validation macro-F1** (best fold: 0.4097). Reviewed per-symptom scores and error patterns; results are experimental, not diagnostic.

Pneumonia Detection Benchmark | Python | PyTorch | CNNs | Transformers | Grad-CAM | [GitHub](#)

- Audited pediatric chest X-rays for corrupt files and exact duplicates, then created balanced validation and test splits.
- Compared five CNN and transformer architectures using a consistent two-phase fine-tuning protocol and validation-selected decision thresholds.
- ConvNeXt-Tiny achieved **0.9997 ROC-AUC and 0.990 recall**; DenseNet121 achieved **0.9899 F1 and 0.995 specificity** on the internal test split. Used Grad-CAM to inspect model attention; results are not clinically validated.

EDUCATION

Central Institute of Technology Kokrajhar

Aug 2021 - Jun 2025

Bachelor of Technology in Computer Science and Engineering | CGPA 8.80/10 | India